Evaluation of steam and water injection effects on gas turbine operation using explicit analytical relations

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Abstract: The changes in performance of a gas turbine resulting from the injection of steam or water into the combustion chamber are evaluated using explicit analytic relations. From the values of performance parameters at an operating point without injection, the changes in fuel amount, power output, and efficiency are evaluated. The difference in behaviour of the case of water and steam injection are highlighted and explained. It is shown that the movement of the operating point on the compressor map, for single and twin shaft turbines, can also be predicted. Comparison with predictions of computer models show that the deviations are predicted accurately. Design data from a large number of commercial gas turbines are used to show the range of variation of parameter deviations and provide some general guidelines as to the size of expected parameter deviations for any gas turbine, when water or steam are injected between the compressor exit and the turbine inlet.

Keywords: gas turbine, steam injection, water injection, performance, efficiency, power output, surge margin

NOTATION

b	compressor external air bleed as a fraction of inlet mass flow
c_p, c_v	specific heat for constant pressure, volume
EGT	exhaust gas temperature
f	fuel/air ratio, $f = \dot{m}_f/\dot{m}_a$
FO	fuel oil
h	specific enthalpy
IGV	inlet guide vane
K	factor accounting for duct and burner pressure losses, $K = \pi_T/\pi_C$
K_b	fractional loss of total pressure in burner
LHV	lower heating value of fuel
$\dot{m}$	mass flow rate
N_c	corrected speed of rotation, $N_c = N/\sqrt{T}$
NG	natural gas
p	pressure (total)
P	gas turbine output power
P_h	specific power, equation (17)
P_T	power produced by turbine
q	reduced mass flow rate, $q = \dot{m}\sqrt{(RT/\gamma)}/p$
r	enthalpy rise of water relative to air, equation (15)
R	specific gas constant
T	temperature (total)
TIT	turbine inlet temperature (also denoted as T_4)
w	(water flow rate)/(fuel flow rate), $w = \dot{m}_w/\dot{m}_f$
war	(water flow rate)/(air flow rate), $war = \dot{m}_w/\dot{m}_a$
γ	isentropic exponent, $\gamma = c_p/c_v$
$\eta_{C_{is}}$	compressor isentropic efficiency
η_{th}	gas turbine thermal efficiency
$\eta_{T_{is}}$	turbine isentropic efficiency
π_C	compressor pressure ratio $\pi_C = p_3/p_2$

Subscripts, superscripts

$'$	value changed because of water injection
0	reference state for gas properties
2,3,4,41,5	position through the gas turbine, Fig. 14.
a	quantity related to air
C	compressor
f	fuel
g	quantity related to the combustion gases
s	quantity related to steam
T	turbine
w	quantity related to injected water

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1 INTRODUCTION

Injection of water or steam into the combustion chamber of gas turbines is one of the means for reducing the emissions of nitric oxides. Steam injection at the compressor outlet is also used as a means of improving performance. If it is produced with the use of the heat of the exhaust gases, it

increases the net efficiency of the gas turbine cycle by exploiting some of the heat that would otherwise be wasted through the exhaust gases. The term diluent will be used hereafter to denote either steam or water injected into the combustion chamber of a gas turbine.

The reduction of nitric oxide emission levels through diluent injection has been the subject of many studies. Reviews of such techniques have been reported [**1**–**3**]. Relations between the amount of water used and the emissions reduction achieved have also been presented [**1**, **4**]. Because steam injection has a beneficial effect on performance, study of its effects on the overall performance of gas turbine cycles has been the subject of many publications [**5**–**8**]. Good summaries of the technology and the achievements up to the early 1990s have been provided [**9**, **10**].

The effects of the injection on the performance of the gas turbine engine are qualitatively known [**11**]. Curves of the dependence of performance parameters on the amount of injected water or steam have been presented by some authors [**12**,**13**]. Such curves are usually produced with the aid of computational performance models of specific gas turbines. The effect of diluent injection is evaluated by introducing the appropriate parameters in a thermodynamic engine model and comparing the results with those having no injection. A study of this type was also reported in [**14**]. The information thus provided is true for a specific gas turbine engine, but nothing is known on the generality of the predictions or about the factors governing the phenomena studied. This means that when another gas turbine is considered, it is not known whether information already available can be used for assessing these effects. In general, a computer model with the capability to handle these phenomena, for the engine of interest, should be used.

These drawbacks do not exist when explicit relations can be used. If such relations are available, then it is possible to study any given gas turbine, provided that the necessary design data are available. Additionally, the use of relations allows an understanding of the effect of diluent injection and provides a means for a quick estimation of such effects. Such relations have been published by the present author [**15**] for studying the effect of water injection on the performance of a gas turbine. In the present paper, this approach is extended to study the effects of steam injection. The analysis is further extended to study the movement of the compressor operating point and thus allow implementation on twin shaft gas turbines. Differences for the effects of steam or water injection will be highlighted and explained. The study is concentrated on the behaviour of the gas turbine engine and it is shown that it can be used to interpret the changes observed on cycle performances.

2 THE PHYSICS OF OPERATIONAL PARAMETER DEVIATIONS

Injection of a diluent between the compressor outlet and the turbine inlet results in the change of several operational parameters. The manner of change depends on the mode of operation. For example, different changes occur if the controller maintains a constant turbine inlet temperature (*TIT*) or if the load demand is kept constant. We will discuss here the changes for operation with given *TIT*, since this is the mode of operation used for defining turbine ratings.

Since the diluent is injected at a temperature that is lower than the *TIT*, some additional heat is needed to bring it to the required *TIT* level. The engine controller therefore gives a command for additional fuel injection into the combustion chamber. The amount of additional fuel depends on the condition of the diluent. When water is injected, the amount of additional fuel is larger than for steam, since the heat needed includes the latent heat of evaporation of the water, which is of considerable magnitude.

Diluent injection results in an increase of the mass flow that has to pass through the turbine. This means that for a given *TIT*, the pressure has to increase (since the referred mass flow rate though the turbine is fixed). Therefore, the compressor is required to operate at a higher pressure ratio. Compressor outlet pressure and temperature are thus altered in comparison to operation with no injection. The movement of the operating point on the compressor map will depend on the shaft configuration, namely single or twin shaft.

The turbine is fed with a larger amount of gas, at a pressure higher than for dry operation. Additionally, this gas has a larger steam content, and therefore a larger heat capacity (the heat capacity of steam is roughly double that of air). This means that it will produce more power. The immediate consequence of these changes is that overall engine performance, namely power output and efficiency, will be altered.

The analysis that follows has as a purpose the evaluation of overall performance changes. The formulae employed have been initially derived by the author in [**15**]. A summary of the derivation procedure is given in Appendix A, for completeness. We start the discussion with the effects on fuel/air ratio and specific heat and continue with the performance parameters.

3 FUEL QUANTITY AND GAS PROPERTIES CHANGES

An important mechanism behind overall performance parameter changes lies in the fact that additional fuel is needed for keeping the turbine inlet temperature constant, as mentioned above. The fuel to air ratio f' when diluent is injected, with a proportion to fuel denoted by w_2 can be evaluated from the ratio f without injection, through the following relation:

$$f' = \frac{f}{1 - rwf} \tag{1}$$

which is derived in Appendix A, equations (14)–(16). Equation (1) can also be written as:

$$\frac{\delta f}{f} = \frac{f' - f}{f} = \frac{rwf}{1 - rwf} \tag{1a}$$

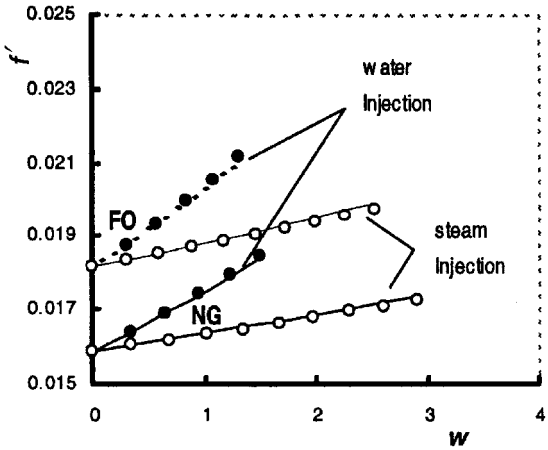

Fig. 1 Typical variation of fuel/air ratio, with water and steam injection (TG20 simulation). Operation with fuel oil and natural gas. Points originate from the computer model, lines are derived from equation (1)

The parameter that determines how f changes with w is the relative enthalpy rise of steam r (equation (15)). This parameter takes characteristic values, which differ for water or steam, and do not change very much for practical gas turbine designs. Its value depends on *TIT*, compressor delivery temperature, and the condition of the injected diluent. It is shown in Appendix B that r does not vary very much for existing gas turbine designs and a typical value that can be used for steam injection is $r \approx 2$ and for water injection $r \approx 5$.

Predictions of fuel/air ratio for different diluent flow rates, using equation (1), are shown in Fig. 1. They are compared to predictions of a detailed thermodynamic computer model. The test case of a single shaft gas turbine, the modelling of which has been reported in [**16**], was used. The predictions of equation (1) are shown almost to coincide with those of the full model, for either steam or water injection. The slope of the change for water is larger than for steam, due to the larger value of r, reflecting the larger amount of fuel needed to evaporate the water, as mentioned in the previous section.

The range of variation of w for water injection is chosen to be narrower than for steam. This choice is made to reflect the fact that water is used mainly for NO_x abatement, and it is more effective than steam for that purpose (it is mentioned in [**2**] that the same level of emissions reduction can be achieved by using steam in quantities 1.6 times that of water). Typically, the maximum reduction can be achieved with a one-to-one proportion of liquid water to fuel. On the other hand, steam may be chosen to be used in larger amounts, not simply for NO_x abatement, but also for performance improvement purposes. For example, a large engine manufacturer uses a typical level of steam content for performance augmentation of the order of 5 per cent of inlet airflow, while recent improvements have allowed the increase of this level to 9 per cent, as reported in [**17**].

3.1 Gas properties

The variation of c_p of the mixture of combustion gases and steam, for the range of temperatures of interest to gas turbine applications, is shown in Fig. 2. The values have been calculated by considering the gases as a mixture of ideal gases, and the properties of the constituents are specified by polynomial relations [**11**]. The variation of isentropic exponent γ is also shown.

The derivation of c_p and γ from the dry combustion gas values can be evaluated by means of a simplified relation. As shown in Appendix A, if property values are known for dry combustion gases, then the change of c_p because of the injection of water or steam can be evaluated though the relation:

$$\frac{\delta c_{pg}}{c_{pg}} = f' w \left(\frac{c_{ps}}{c_{pg}} - 1 \right) \tag{2}$$

The specific heat of the dry combustion products can be considered to be the same as the one before the diluent is

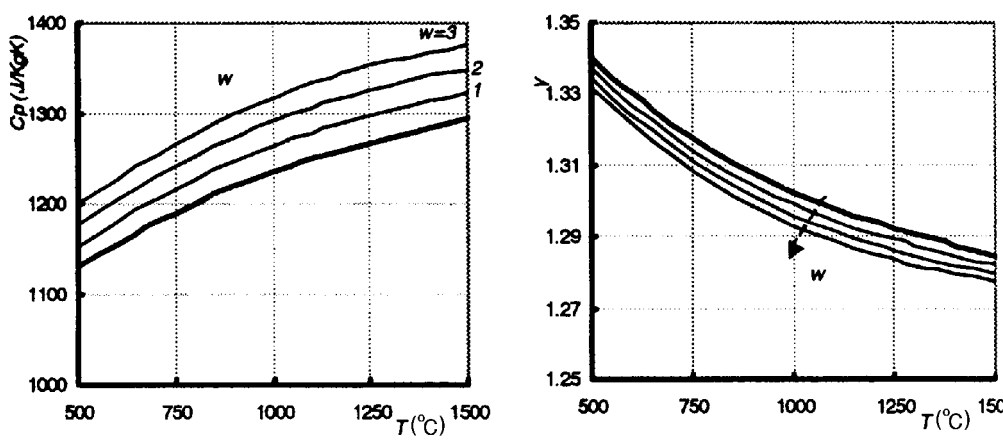

Fig. 2 Values of specific heat C_p and isentropic exponent γ, for temperatures usual in the hot section of gas turbines

injected. A similar relation can be derived for the change in the value of the gas constant R:

$$\frac{\delta R_g}{R_g} = f'w\left(\frac{R_s}{R_g} - 1\right) \tag{3}$$

The change in the isentropic exponent γ, from its definition and the previous expressions, is obtained as:

$$\frac{\delta\gamma}{\gamma} = (\gamma - 1)\left(\frac{\delta R_g}{R_g} - \frac{\delta c_{pg}}{c_{pg}}\right) = (\gamma - 1)\left(\frac{R_s}{R_g} - \frac{c_{ps}}{c_{pg}}\right)f'w \tag{4}$$

The product $(f'w)$ appearing in these relations is equal to the water-to-air ratio *war*. If this quantity is substituted in the previous expressions, they could also be used for changes in inlet air properties due to varying ambient humidity.

We note that the above expressions can be further simplified for gas turbine combustion gases, by taking into account that the specific heat of steam is roughly double the specific heat of the combustion gases, as can be seen from Fig. 3. For the range of turbine inlet temperatures encountered in today's turbines, and representative values of fuel/air ratio, c_{ps}/c_{pg} is shown to take values very close to 2.

By virtue of this observation, equation (2) for c_p becomes:

$$\frac{\delta c_{pg}}{c_{pg}} \approx f'w \tag{5a}$$

For the gas constant R, for $R_s = 461.5$ and $R_g = 287$, equation (3) gives:

$$\frac{\delta R_g}{R_g} = 0.61f'w \tag{5b}$$

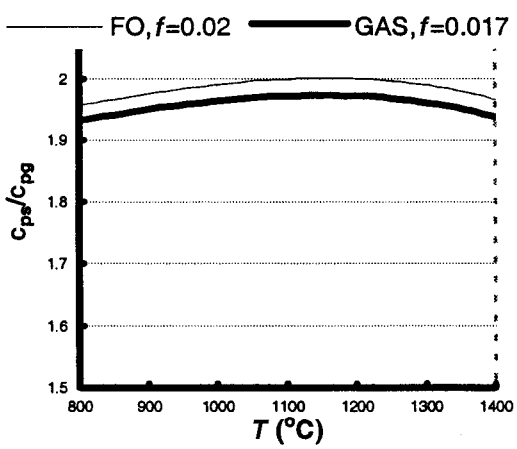

Fig. 3 Ratio of specific heats of steam and combustion gases

For the isentropic exponent, using these relations and equation (4):

$$\frac{\delta\gamma}{\gamma} \approx 0.33(1.61 - 2)f'w = -0.13f'w \tag{5c}$$

4 PERFORMANCE DEVIATIONS

With the help of the formulae for the fuel/air ratio and specific heat deviations, simplified expressions for the deviations of power output and efficiency can be derived. We will show that use of such expressions allows the evaluation of the effects of diluent injection on the performance of a gas turbine. Power output and efficiency are examined first, followed by the effect on the compressor operating point and exhaust gas temperature.

4.1 Power output

As shown by equations (17)–(19) in Appendix A, the fractional change in power output is given by the following approximate relation:

$$\frac{\Delta P}{P} = \frac{\Delta\dot{m}_2}{\dot{m}_2} + 2\frac{P_T}{P}f'w \tag{6}$$

The first term is the fractional change of air mass flow rate caused by the change in the operating point of the compressor. The second term represents the change in power produced by the turbine, swallowing extra fluid, namely the injected diluent. The factor 2 is derived from the fact that steam produces twice the power produced by the same mass flow rate of combustion gas, when expanding over the same temperature limits, due to its thermal capacity being double that of the combustion gas. Equation (6) can be modified to use only data from the dry operating point at the same *TIT*, by substituting f' from equation (1):

$$\frac{\Delta P}{P} = \frac{\Delta\dot{m}_2}{\dot{m}_2} + 2\frac{P_T}{P}\frac{fw}{1 - rwf} \tag{6a}$$

For a modern single shaft turbine used in electricity generation, operation with fixed compressor geometry implies almost no change in inlet mass flow rate. Therefore, the change in power will depend only on the water/fuel ratio w, since the first term on the right-hand side of equation (6) will be almost zero. When variable geometry exists and is used for engine control, mass flow changes will have to be accounted for according to this relation. This also holds for a twin shaft gas turbine.

Variation of the power output of a single shaft gas turbine with constant IGV setting, as a function of the water-to-fuel ratio, is shown in Fig. 4. Results for operation with fuel oil or natural gas and injection of water or steam are shown. The lines represent the prediction using equation (6) and the

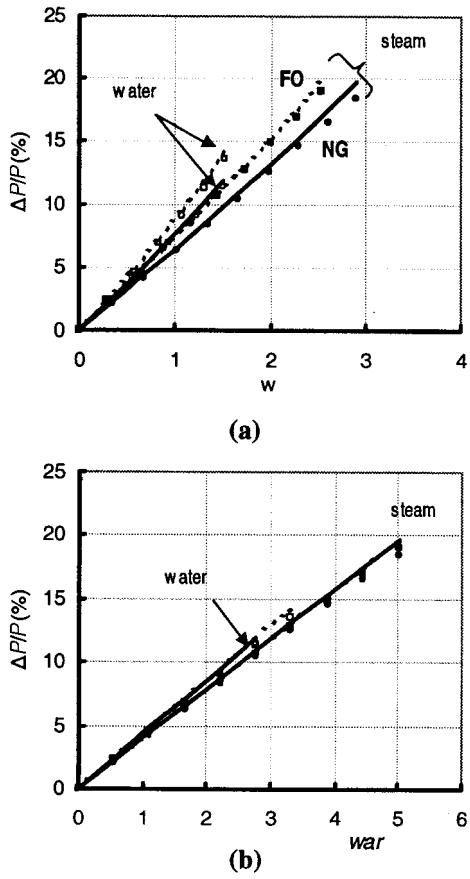

Fig. 4 Power output deviation with diluent injection (a) as a function of the water/fuel ratio, (b) as a function of the water/air ratio. Points originate from the computer model, lines are derived from equation (6)

points prediction of the same conditions with a detailed thermodynamic computer model of the engine considered [**16**]. It is shown that predictions with equation (6) are in very close agreement with the predictions of the full model, and thus equation (6) can accurately represent changes due to diluent injection for different fuels. It can also be used to interpret the differences in operation for the different cases, shown in this figure.

Water injection produces a larger power output than the same quantity of steam, irrespective of the kind of fuel. Equation (6a) can be used to explain this trend: for a given operating point and fuel (given P_T/P) the slope of the power increase versus w is higher in the case of water, since the parameter r takes a larger value, and thus the denominator in equation (6a) becomes smaller. It is also observed in Fig. 4 that the change in power output is smaller when the fuel is gas, as compared to liquid. The reason for this behaviour is twofold: first, the term P_T/P is slightly smaller (P_T increases due to a larger gas specific heat and therefore $P_T/P[=1/(1-P_C/P_T)]$ slightly decreases), and secondly, the slope is proportional to f, which is smaller for a typical natural gas (due to its LHV being about 20 per cent higher than fuel oil). The difference is much smaller if war $(=f'w)$ is plotted on the abscissa, as shown in Fig. 4b. The use of this independent variable thus virtually eliminates the dependence on the type of fuel.

The values of the predictions presented in Fig. 4 are in very close agreement with data published by gas turbine manufacturers, as for example those shown in Fig. 5. Identical data were presented in [**12**]. It is interesting to note that although the design point data of the engines referred to in the published cases are not mentioned, the agreement with the predictions of the present method is very close. A reason for this will become apparent later, in the discussion section of the paper.

4.2 Efficiency

As shown by equations (20) and (21) of Appendix A, the deviation of thermal efficiency is given by the relation:

$$\frac{\Delta(\eta_{\text{th}})}{\eta_{\text{tth}}} = \left(\frac{2P_T}{P} - r\right) f'w \tag{7}$$

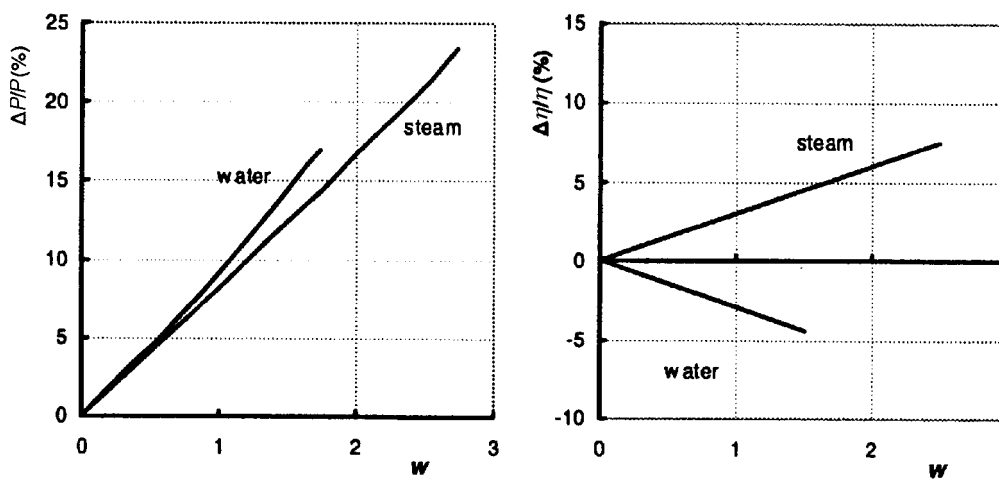

Fig. 5 Data published by gas turbine manufacturers. Data from [**13**]

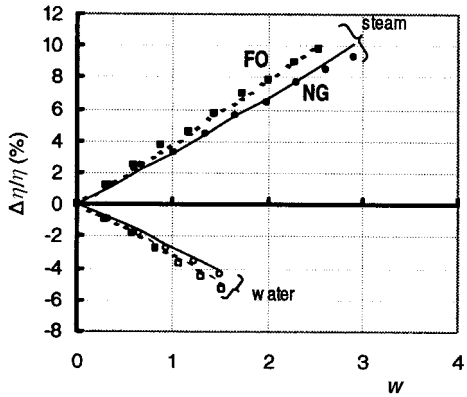

Fig. 6 Gas turbine efficiency deviation with diluent injection

which can be further transformed, using equation (1)

$$\frac{\Delta(\eta_{\text{th}})}{\eta_{\text{th}}} = \left(\frac{2P_{\text{T}}}{P} - r\right)\frac{fw}{1 - rfw} \tag{7a}$$

The efficiency deviation, as a function of the water-to-fuel ratio, is shown in Fig. 6, where data for water injection and different fuels are also included, similar to Fig. 4. The good agreement with predictions of the full computer model can be observed in this case too, while the trends predicted here are close to the trends published by engine manufacturers, shown in Fig. 5. In this case, too, it can be shown that use of the water-to-air ratio as an independent variable produces curves that demonstrate a very weak dependence on the kind of fuel.

The first observation is that while both steam and water injection result in an output increase, they have the opposite effects on efficiency: steam injection increases efficiency while water injection decreases it. The reason for this behaviour can be explained by looking at equation (7). As discussed above, r is very large for water injection, and thus the coefficient in equation (7) becomes negative. It has a much smaller value for steam and hence the positive slope.

The change in efficiency can be further interpreted physically, as the resultant of two effects in opposite directions: more power is produced due to the increased flow through the turbine and increase in heat capacity. This power increase is achieved with an increase in the fuel consumed, to heat the diluent injected. This heat is proportionally higher than the extra power, in the case of water injection, as it includes the heat of vaporization. This proportionality is expressed by the value of r. For water r is very large, and offsets the advantage in power. It should be noted that this effect can also be seen from the cycle point of view: in the case of steam the average temperature of heat addition to the cycle is slightly increased, while the temperature of heat rejection is lowered, since part of the exhaust heat is used to produce the steam. In the case of water, the heat rejection is at the same temperature, while heat addition occurs at a lower temperature (a large part of it is used to evaporate the water).

4.3 Compressor operating conditions

Movement of the operating point on the compressor map is of interest for different operating conditions, mainly to monitor surge margins and to ensure that the risk of encountering unstable operation is minimized. The fact that steam injection reduces surge margin means that compressor operation has to be monitored, to avoid the occurrence of such conditions, as discussed in [**18**].

The operating point of the compressor is defined by the pressure ratio π_{C} and reduced mass flow rate q_2. Applying equation (22) of Appendix A, for operation without and with water injection and dividing the two equations gives:

$$\frac{\pi'_{\text{C}}}{\pi_{\text{C}}} = \frac{q'_2}{q_2}\frac{1 + f'(1 + w)}{1 + f}\sqrt{\frac{R'_4}{R_4}\frac{\gamma_4}{\gamma'_4}\frac{T'_4}{T_4}\frac{T_2}{T'_2}} \tag{8}$$

This relation can be used to predict the pressure ratio at any operating condition once the pressure ratio at one operating point is known (e.g., at base load) and can be applied to both dry and wet operation. It can be used to estimate the change in pressure ratio, which will result from the injection of water. For constant *TIT*, this equation gives:

$$\frac{\pi'_{\text{C}}}{\pi_{\text{C}}} = \frac{q'_2}{q_2}\frac{1 + f'(1 + w)}{1 + f}\sqrt{\frac{R'_4}{R_4}\frac{\gamma_4}{\gamma'_4}} \tag{9}$$

Using the small $f'w$ approximation and equation (23), it can be transformed as follows:

$$\frac{\pi'_{\text{C}}}{\pi_{\text{C}}}\frac{q_2}{q'_2} \approx (1 + f'w)(1 + 0.37f'w) \approx 1 + 1.37\frac{fw}{1 - rwf} \tag{10}$$

At this point it must be noted that operation of a single shaft configuration and a configuration with a free power turbine is different. The relations for change in pressure ratio, presented above, can be used to estimate the shift of operating points for the two cases.

4.3.1 *Single shaft gas turbine*

The typical application for this configuration is electricity generation. In such applications the shaft mechanical speed is constant. This means that for given ambient conditions the compressor operating point will be moving along a constant speed line. For operation with constant geometry, namely with no variation of stator vane angles, it is usual that the nominal constant speed characteristic is approximately vertical to the flow axis, namely $q_2 =$ constant, especially for high pressure ratio transonic axial compressors. Opera-

tion at constant turbine inlet temperature thus allows a further simplification of equation (9):

$$\frac{\pi'_C}{\pi_C} = \frac{1+f'(1+w)}{1+f}\sqrt{\frac{R'_4}{R_4}\frac{\gamma_4}{\gamma'_4}} \approx (1+f'w)\sqrt{\frac{R'_4}{R_4}\frac{\gamma_4}{\gamma'_4}} \qquad (11)$$

This relation shows that pressure ratio changes as a result of two effects: change of mass flow though the turbine and change of gas properties as a result of increased water content. Therefore, diluent injection will force the pressure ratio to increase.

Equation (11) can be used to evaluate change in pressure ratio and thus estimate the new position of the compressor operating point. An example of a calculated change in pressure ratio is shown in Fig. 7, where it is also shown that predictions of equation (11) are in close agreement to those of the full computer model. The way the operating point moves is shown in Fig. 8.

4.3.2 *Twin shaft gas turbine*

For dry operation, the gas generator operating point moves along the operating line. For a single spool gas generator, the movement of the operating point can be assessed by using the power balance equation. This equation can be written as follows:

$$\pi_C^{(\gamma-1)/\gamma} - 1 = \frac{m_4}{m_2}\frac{c_{pg}}{c_{pa}}\frac{T_4}{T_2}\left(1 - \frac{T_{41}}{T_4}\right)\eta_{C_{is}}$$

For operation with the power turbine choked, the temperature ratio of the gas generator turbine T_{41}/T_4 remains approximately constant [**23**]. If we consider operation with a given T_4, write this equation for operation with and without injection, divide the two equations and use also equation (18), we obtain:

$$\frac{\pi_C'^{\gamma-1/\gamma} - 1}{\pi_C^{\gamma-1/\gamma} - 1} = \frac{1+f'(1+w)}{1+f}\frac{c'_{pg}}{c_{pg}} \qquad (12)$$

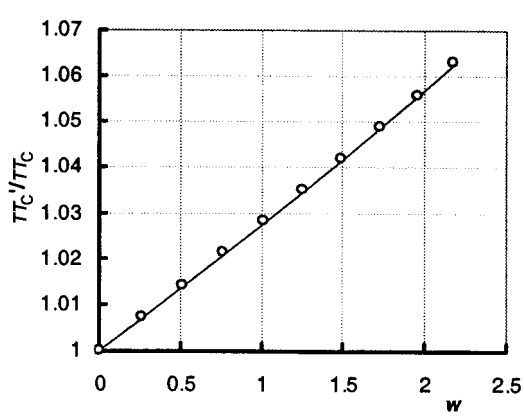

Fig. 7 Variation of compressor pressure ratio as a function of the amount of injected steam for a single shaft gas turbine

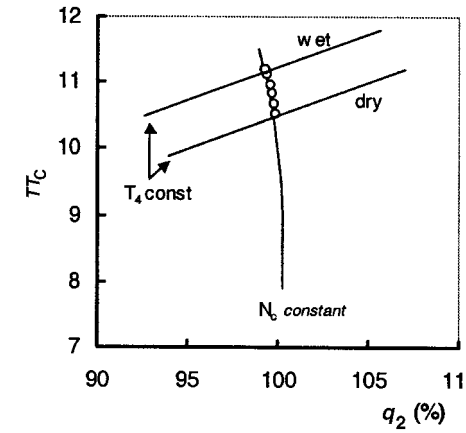

Fig. 8 Change of compressor operating conditions with diluent injection for a single shaft gas turbine

This equation can be used to predict the change in pressure ratio. A comparison of prediction of this equation and prediction using a full computer model of a twin shaft gas turbine (the one described in [**19**]), is shown in Fig. 9.

Once the pressure ratio is estimated, the compressor flow function q_2 can be estimated using equation (9). The movement of the operating point on the compressor map, corresponding to the predictions of Fig. 9, is shown in Fig. 10, where predictions of this equation are compared to those of the full computer model. It is observed that pressure ratio varies with mass flow rate and compressor speed. The increase in power caused by diluent injection drives the compressor to operate at higher rotational speed; the surge margin is, however, reduced.

The estimated deviations of the compressor flow function provide directly the deviations of mass flow rate from the value at the dry operating point. This means that both terms of equation (6) are known and thus the deviation in power output can be assessed for the case of the twin shaft gas turbine as well. A comparison of the deviation evaluated

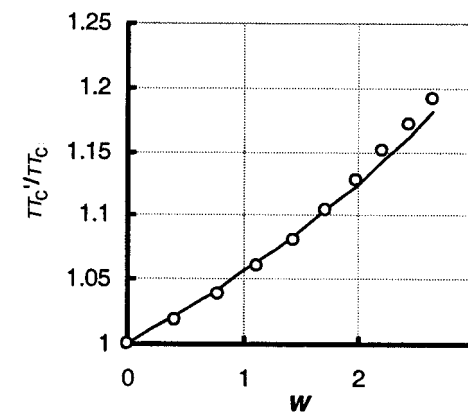

Fig. 9 Change of compressor pressure ratio with water injection for a twin shaft gas turbine. Points originate from the computer model, line is derived from equation (12)

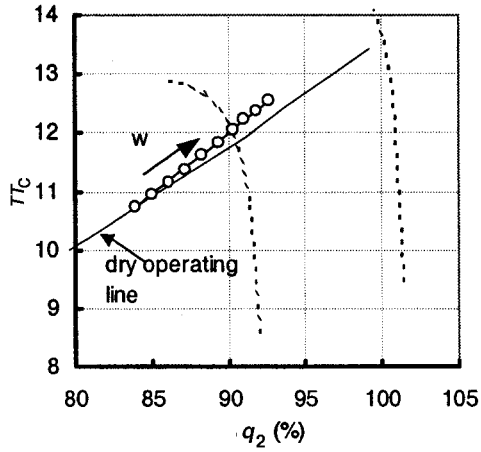

Fig. 10 Change of compressor operating conditions with diluent injection for a twin shaft gas turbine. °, operating points when water is injected at constant *TIT*

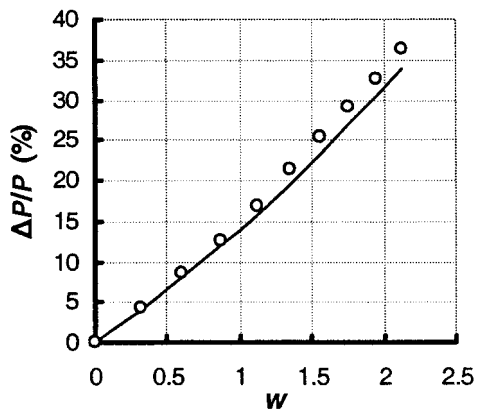

Fig. 11 Deviation of power output with water injection for a twin shaft gas turbine

through equation (6a) and the detailed computer model is shown in Fig. 11.

5 DISCUSSION

All the relations presented above for evaluating the deviations of performance parameters contain the term $f'w$. This term is equal to the water/air ratio, since:

$$f'w = \frac{\dot{m}_f}{\dot{m}_a}\frac{\dot{m}_w}{\dot{m}_f} = \frac{\dot{m}_w}{\dot{m}_a} = war$$

This means that this parameter can also be considered as the governing parameter for the deviations. We have also shown that when this parameter is used, the change in power output and efficiency depends very little on the type of fuel, while when w is the independent variable, different slopes are observed for different fuels. On the other hand, the use of w may be preferable since this parameter is the one that determines the reduction in NO_x emissions, as discussed in the introduction.

The formulae for power output and efficiency are valid, independently of shaft arrangement. It was discussed previously that in the case of a single shaft gas turbine with constant geometry, the amount of water alone determines the power change. If variable IGVs are used their control law would need to be known, for evaluating the power change (this law would allow determination of the $\Delta\dot{m}/\dot{m}$ in equation (6). For the twin shaft gas turbine, the mass flow change is evaluated from the power and flow matching relations, as discussed in the previous section.

It is interesting to examine how the sensitivity of power and efficiency variations are expected to vary for the range of design parameters encountered in today's gas turbines. The main parameters that influence sensitivity to diluent injection are the ratio of turbine power to power output P_T/P and the relative enthalpy rise of the steam expressed by the parameter r. The range of values of these two parameters for today's gas turbines is discussed in Appendix B. On the basis of these values, the range of variation of the power versus water-to-air ratio for constant geometry single shaft gas turbines, is shown in Fig. 12. *war* is chosen as the abscissa, since the corresponding lines are then almost independent of the fuel type. It is observed that for a large variety of designs, the rate of power change with water-to-air ratio varies within a relatively narrow range. The range for water injection is also narrow with slightly higher slope. From this figure we could claim that for any gas turbine, a rule of thumb would be that there is an increase of 5 per cent in power for every 1 per cent of injected water or steam, expressed as a per centage of inlet air, for operation at nominal conditions.

At this point it should be noted that the increase in output for a twin shaft gas turbine is larger, for the same amount of injected water or steam. For this case, diluent injection results in an increase of mass flow rate, resulting in an

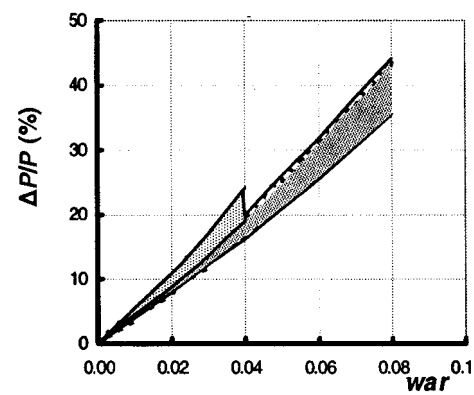

Fig. 12 Range of variation of power deviation for existing gas turbines (single shaft, constant geometry)

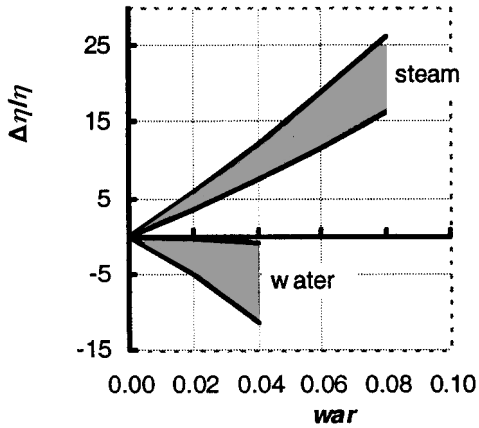

Fig. 13 Range of variation of efficiency deviation for existing gas turbines (single shaft, constant geometry)

additional power increase, as indicated from equation (6). The simultaneous increase in pressure ratio results in general in larger values of the ratio P_T/P, giving finally a larger increase of output per unit diluent mass flow injection. Analysis of equations (10) and (12) can show that for this case an additional output increase of about 1.5 per cent per 1 per cent of injected diluent should be expected. We note here that in [**14**], an increase of 6.5 per cent per 1 per cent of injected steam in a twin spool, twin shaft gas turbine was reported, which is in agreement with the previous observations.

The corresponding changes of efficiency are shown in Fig. 13. It is observed here that the efficiency increase due to steam injection is bounded within a relatively narrow band. The range of variation for the efficiency drop for water injection is rather wide however.

Both Figs 12 and 13 have been derived on the basis of design point data. When off-design operation of a given gas turbine is considered, the curves may differ considerably, since the parameter P_T/P may change.

In the analysis presented above the relations for fuel/air ratio, power output, and efficiency have been derived for constant *TIT*. This is the usual way performance parameter deviations are evaluated [**8**]. We point out, however, that this mode of operation is not necessarily the one implemented in the control system of a gas turbine. It is known (as for example discussed in [**21**]) that the control mode may be such that the turbine inlet temperature is lowered when water is injected. This is done because the increase in heat transfer coefficient due to the increased steam content results in a higher blade temperature for the same gas temperature. If it is desired to keep the blade temperature constant, the gas temperature is lowered. The term 'dry' control mode is used in this case, as opposed to 'wet' control mode, for which turbine inlet temperature is kept constant. The analysis presented here can also be used for that case. Deviations due to diluent injection can be superimposed on those due to *TIT* variation, as long as both represent a relatively small percentage of the overall values.

The present analysis can also be useful in practical situations, when correction curves are provided by the engine manufacturer, accounting for the amount of water or steam injected. Such curves may be used to refer data from performance testing to a certain amount of water injected. The formulae given above can be used for a check of the consistency of the curves provided with the gas turbine design data. They could even be used as a basis for correction, if needed.

6 CONCLUSIONS

The possibility of predicting the main gas turbine performance parameter deviations due to the injection of water or steam into the combustion chamber by means of simple analytical relations has been demonstrated. It was shown that with minimum design data, it is possible to predict deviations in performance with an accuracy comparable to that of more detailed models.

These relations allow an interpretation of the difference in behaviour observed between water and steam injection, or even different fuels. It was thus explained why both water and steam injection have the same influence on output, while they produce opposite trends in efficiency.

Apart from the overall performance, it was shown that the movement of the operating point on the compressor characteristic can also be predicted.

The generality of the predictions was discussed and it was shown that some general trends exist for current gas turbine designs, as demonstrated by using published design data from a large number of industrial gas turbines.

It was also shown that the main parameters influencing the magnitude of the deviations caused by diluent injection is bounded within certain limits. Some generalized conclusions can thus be derived for any application.

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APPENDIX A

The layout of a gas turbine considered for the present study is shown in Fig. 14. This layout covers both types of shaft arrangement. For the twin shaft gas turbine, station 41 is the station between the compressor and the power turbine.

The formulae used to calculate performance parameter changes have been derived by the author in [**15**] for water injection. The approach followed is to consider an operating point without diluent injection and to derive formulae for the deviations of performances from the values at that operating point. From the derivation, it is deduced that they are also applicable to steam injection. Injection at the compressor outlet or within the combustion chamber are equivalent from a performance point of view.

Derivation of simplified relations becomes possible by taking advantage of the fact that the fuel-to-air ratio f of a gas turbine is a small number, with a value usually around 0.02 for base load operation. Terms containing a higher power of this quantity can thus be neglected without loss of accuracy. A summary of the procedure for derivation of the equations is given in the following.

Gas properties

Diluent injection increases the steam content of the hot gases into the turbine in comparison with the case with no diluent injection. The new specific heat can be calculated if the gas is considered to be a mixture of water vapour and gas produced from dry combustion:

$$c'_{\mathrm{pg}} = \frac{(1+f')c_{\mathrm{pg}} + f'wc_{\mathrm{ps}}}{1+f'(1+w)}$$

Taking into account that f', $f'w \ll 1$ this relation can be simplified to give:

$$\frac{c'_{\mathrm{pg}}}{c_{\mathrm{pg}}} \approx \left[1 + f'w\left(\frac{c_{\mathrm{ps}}}{c_{\mathrm{pg}}} - 1\right)\right] \tag{13}$$

From this relation, equation (2) of the main text is derived.

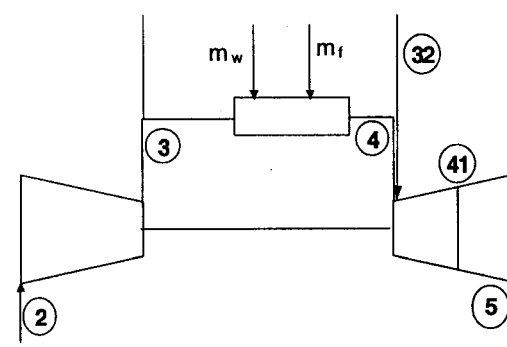

Fig. 14 Schematic of the gas turbine layout studied

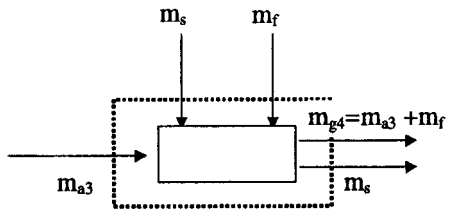

Fig. 15 Control volume for application of the SFEE over the combustion chamber

Fuel-to-air ratio

The change in fuel flow rate when diluent is injected can be evaluated by applying the steady flow energy equation (SFEE) to the combustion chamber for operation with and without injection. The flows into and out of the combustion chamber are shown schematically in Fig. 15.

The SFEE for the combustion chamber is:

$$(1+f')(h_{g4}-h_{g0}) = (h'_{a3}-h_{a0})+f'LHV-wf'(h_{s4}-h_w)$$

Writing this equation for dry and wet operation and eliminating the quantity *LHV*, gives the following relation for fuel-to-air ratio:

$$\frac{f'}{f} = rwf' + a \tag{14}$$

where

$$a = \frac{(h_{g4}-h_{g0})-(h'_{a3}-h_{a0})}{(h_{g4}-h_{g0})-(h_{a3}-h_{a0})}, \quad r = \frac{h_{s4}-h_w}{(h_{g4}-h_{g0})-(h_{a3}-h_{a0})} \tag{15}$$

The value of a in this relation is very close to unity. Compressor outlet enthalpy h'_{a3} differs from the dry value h_{a3} by a relatively small amount, compared to the enthalpy rise in the combustion chamber. The numerator and denominator of the fraction are therefore almost equal. We thus have:

$$\frac{f'}{f} \approx rwf' + 1 \tag{16}$$

which gives equation (1) of the main text.

We note that r expresses the relative specific enthalpy rise of the injected diluent with respect to the enthalpy rise for transformation of the air into combustion products.

Power output

The deviation of power output has been derived in [**15**] and is also summarized here for completeness. Power output is derived by relating the net output to the difference between the turbine and compressor power:

$$P = \dot{m}_2 c_{pa} T_2 \times \overbrace{\left[\frac{\dot{m}_4}{\dot{m}_2}\frac{c_{pg}}{c_{pa}}\frac{T_4}{T_2}\eta_{T_{is}}\left(1-\frac{1}{\pi_T^{\gamma_g-1/\gamma_g}}\right)-\frac{1}{\eta_{C_{is}}}(\pi_C^{\gamma_a-1/\gamma_a}-1)\right]}^{P_h} \tag{17}$$

This is a simplified expression, as it does not take into account the effect of bleed flows, such as those used for cooling. It provides a better approximation when T_4 is the ISO turbine inlet temperature (derived on the assumption of mixing bleed flows at the turbine inlet). Deviations, when water or steam is injected, are calculated more accurately, provided that the cooling flows remain unchanged. This is a good approximation, since these flows are driven by pressure differences that remain practically unaltered.

For given ambient temperature and turbine inlet temperature, taking logarithms and differentiating, we obtain:

$$\frac{\delta P}{P} = \frac{\delta \dot{m}_2}{\dot{m}_2} + \frac{m_2 c_{pa} T_2}{P}\delta P_h \tag{18}$$

where P_h is the term in brackets in equation (17). The deviation δP_h, can be further evaluated as follows:

$$\delta P_h = Y_T \cdot \delta\left(\frac{\dot{m}_4}{\dot{m}_2}\frac{c_{pg}}{c_{pa}}\frac{T_4}{T_2}\right) + \left(\frac{\dot{m}_4}{\dot{m}_2}\frac{c_{pg}}{c_{pa}}\frac{T_4}{T_2}\delta Y_T - \delta Y_c\right) \tag{18a}$$

where

$$Y_C = \pi_C^{(\gamma_a-1)/\gamma_a} - 1/\eta_{C_{is}}, \; Y_T = \eta_{t_{is}}\left(1 - 1/\pi_T^{\gamma_g-1/\gamma_g}\right)$$

For evaluation of the first term of the right-hand side of equation (18a), when ambient and turbine inlet temperature are constant, we have:

$$\delta\left(\frac{\dot{m}_4}{\dot{m}_2}\frac{c_{pg}}{c_{pa}}\right) = \frac{\dot{m}_4}{\dot{m}_2}\frac{c_{pg}}{c_{pa}}\left(\frac{\delta(\dot{m}_4/\dot{m}_2)}{\dot{m}_4/\dot{m}_2} + \frac{\delta c_{pg}}{c_{pg}}\right) \tag{18b}$$

The mass balance for the combustion chamber gives the following relation between the turbine inlet and compressor inlet mass flow rates:

$$\frac{\dot{m}_4}{\dot{m}_2} = \frac{\dot{m}_4}{\dot{m}_3}\frac{\dot{m}_3}{\dot{m}_2} = [1 = f'(1+w)]\cdot(1-b) \tag{18c}$$

where b is the fraction of the total air flow bled from the compressor before entry to the combustion chamber.

Application of this equation without and with water injection, gives:

$$\delta\left(\frac{\dot{m}_4}{\dot{m}_2}\right)\Big/\left(\frac{\dot{m}_4}{\dot{m}_2}\right)=\frac{fw}{1+f} \tag{18d}$$

The second term of the right-hand side of equation (18a) expresses the rate of change of specific power with pressure ratio, for given isentropic efficiencies of compressor and turbine. This term is very small and can be neglected for the following reason: gas turbine designers usually choose the design point of an engine in the 'flat' region of the specific-power versus pressure ratio curve, where the variation with pressure ratio is very small.

Using now equations (18a,b,d) and (2), equation (18) gives:

$$\frac{\delta P}{P}=\frac{\delta\dot{m}_2}{\dot{m}_2}+\frac{P_{\mathrm{T}}}{P}\left(\frac{c_{\mathrm{ps}}}{c_{\mathrm{pg}}}-\frac{f}{1+f}\right)f'w \tag{19}$$

For $c_{\mathrm{ps}}/c_{\mathrm{pg}}\approx 2$ and $f\ll 1$, this relation gives equation (6) of the main text.

Efficiency

The relation for estimating efficiency deviation can be derived from the equation of the definition of gas turbine efficiency:

$$\eta_{\mathrm{th}}=\frac{P}{\dot{m}_{\mathrm{f}}LHV}=\frac{P}{f\dot{m}_3LHV} \tag{20}$$

Efficiency change can be derived from this relation, by taking logarithms and then differentiating:

$$\frac{\delta(\eta_{\mathrm{th}})}{\eta_{\mathrm{th}}}=\frac{\delta P}{P}-\frac{\delta f}{f}-\frac{\delta\dot{m}_3}{\dot{m}_3} \tag{21}$$

Substitution of $\delta P/P$ from equation (19) and using also equation (16), produces equation (7) of the main text.

It is noted here that the gas turbine alone is considered in the definition of efficiency. When steam is injected, the heat of producing it is not accounted for in this expression. This represents the case that steam is produced from the gas turbine exhaust gases and no additional fuel is used.

Compressor pressure ratio

Pressure ratio can be related to turbine and compressor flow capacities through flow matching relations. Using the definition of turbine and compressor reduced mass flow rates and equation (18c), which links the inlet mass flow rate to the two components, it is easily deduced that the following relation holds:

$$\pi_{\mathrm{C}}=\frac{q_2}{q_4}(1+f'(1+w))(1-b)(1-K_{\mathrm{b}})\sqrt{\frac{R_4}{R_2}\frac{\gamma_2}{\gamma_4}}\sqrt{\frac{T_4}{T_2}} \tag{22}$$

It should be noted that a generalized expression for the reduced mass flow rates is used, to account for properties alteration due to water addition [**20**]. This relation holds for any operating condition, with or without diluent injection. It is also realistic to assume that the bleed air fraction b and loss factor K_{b} do not change with diluent injection, namely the term $(1-b)(1-K_{\mathrm{b}})$ remains constant.

It is also noted that the turbine flow function q_4 remains constant for a choked turbine, which is the case for the largest part of the operating envelope of modern gas turbines.

Using equation (5) for gas constant and isentropic exponent deviations we obtain:

$$\frac{R_4'}{R_4}\frac{\gamma_4}{\gamma_4'}=\frac{(1+\delta R/R)}{(1+\delta\gamma/\gamma)}=\frac{(1+0.61f'w)}{(1-0.13f'w)}$$
$$\approx 1+0.74f'w\Rightarrow\sqrt{\frac{R_4'}{R_4}\frac{\gamma_4}{\gamma_4'}}\cong 1+0.37f'w \tag{23}$$

APPENDIX B

Design point data for gas turbines

The variation of parameters r and P_{T}/P, which determine the rate of change of power output and efficiency due to diluent injection, is now examined for current gas turbines.

The choice of design point data for gas turbines is dictated by optimization requirements, usually for maximizing specific power or efficiency. The consequence of this is that the design point data for existing gas turbines follow some trend, related to this fact. The two key design parameters for a large number of gas turbines are plotted in Fig. 16. It is observed that the points are concentrated in a

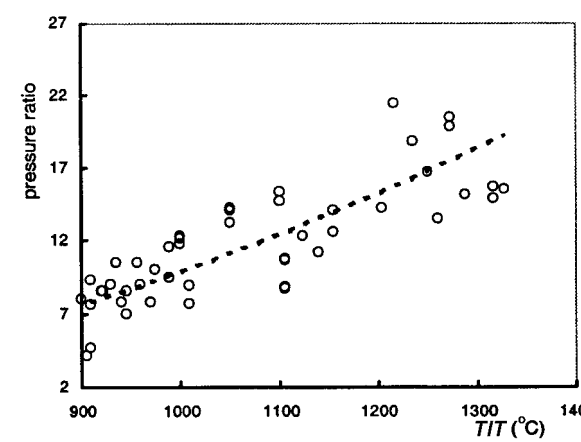

Fig. 16 Nominal pressure ratio and turbine inlet temperature for 89 industrial gas turbines. Data from [**21**, **22**]

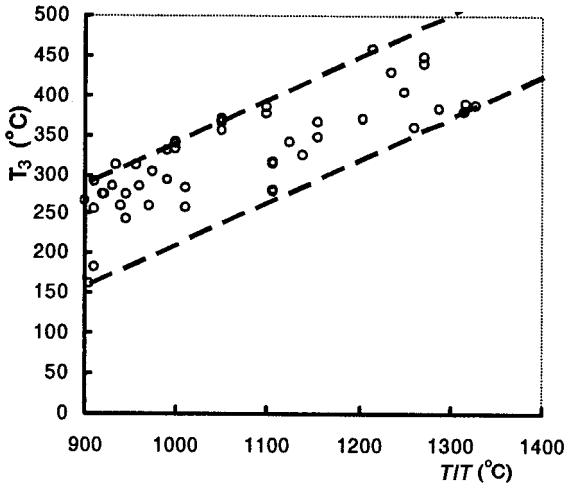

Fig. 17 Compressor delivery temperature for the gas turbines of Fig. 16

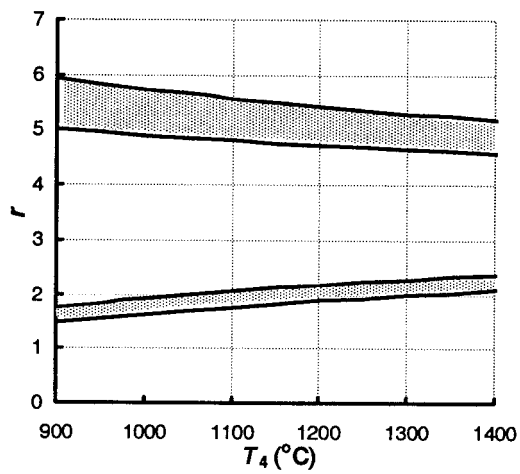

Fig. 18 Values of parameter r for typical cycle variables of gas turbines. The bands correspond to those of Fig. 17

band, as would be expected on the basis of the above argument (if all engines had the same component efficiencies and optimization was carried out according to the same criterion, then this band would have been much narrower).

The temperature at the exit of the compressor for the turbines of Fig. 16 can be calculated for standard day inlet, by assuming a typical value of compressor isentropic efficiency. Values thus calculated are shown in Fig. 17.

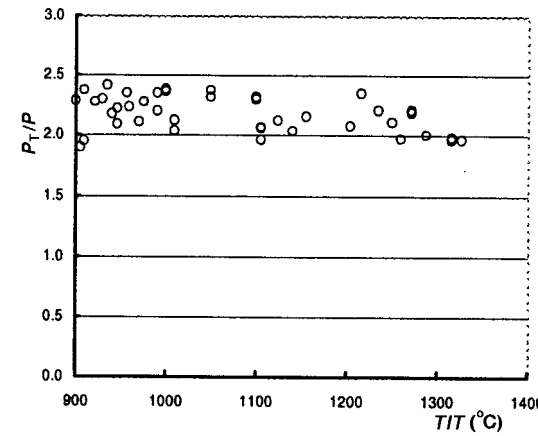

Fig. 19 P_T/P for the gas turbines of Fig. 16

Using now the limit values for compressor delivery temperature, for each turbine inlet temperature the relative steam enthalpy rise r is calculated, and the result is shown in Fig. 18. It is noticed that the values of r do not change significantly for each diluent over the values encountered in current designs. The values for injecting water are significantly larger than those for injecting steam. This is a consequence of the fact that a significant amount of heat is needed for its evaporation.

The ratio of turbine power to output power P_T/P for the gas turbines of Fig. 16 can be calculated by assuming some typical values for component efficiencies, through the following relation:

$$\frac{P_T}{P} = \left[1 - \frac{\dot{m}_4}{\dot{m}_2}\frac{c_{pg}}{c_{pa}}\frac{T_4}{T_2}\frac{1}{\eta_{T_{is}}\eta_{Cis}}\frac{(\pi_C^{\gamma_a - 1/\gamma_a} - 1)}{(1 - (K\pi_C)^{\gamma_g - 1/\gamma_g})}\right]^{-1} \tag{24}$$

The results of the calculation are shown in Fig. 19. Although component efficiencies have been taken to be the same and parasitic flow schemes are not necessarily identical for different turbines, the calculated values can be considered to be a good approximation of actual values. The ISO turbine inlet temperature has been used, which is derived on the assumption of all bleed flows being mixed at the inlet to the turbine. We note that the power ratio shows a small variation and lies in a band between the values 1.9 and 2.4.